k-means Clustering Analysis Report

data.npy (468×9) and data2.npy (493×13) | k range: 1–15

(a) Selection of k

The elbow method selects the value of k at which the marginal reduction in the within-cluster sum of squared errors (SSE) ceases to be substantial. Geometrically, the algorithm draws the chord between the endpoints (k = 1) and (k = 15) and identifies the data point whose perpendicular distance from this chord is maximised. Figure 1 shows the SSE curves for both datasets, with dashed vertical lines marking the geometric elbow for each.

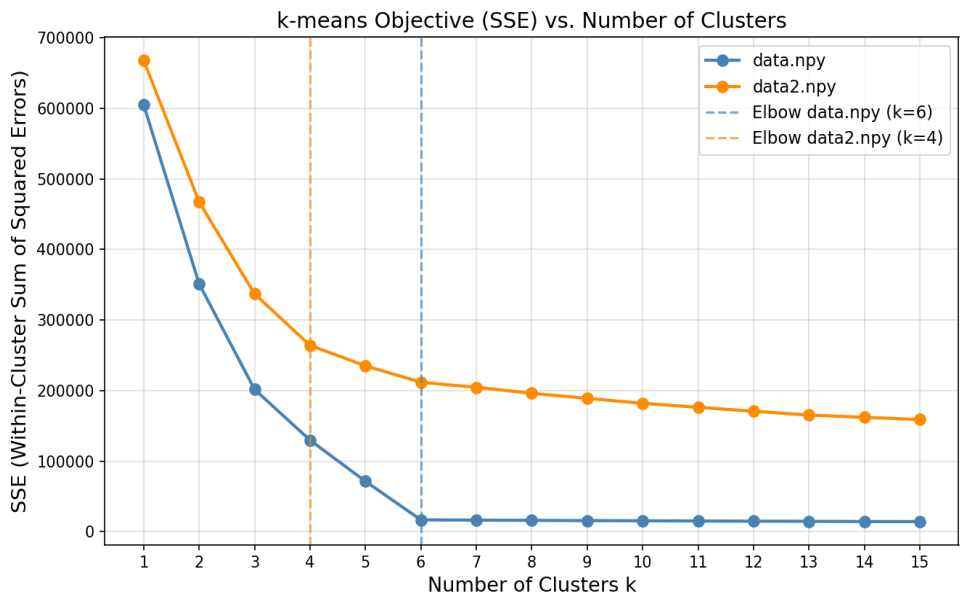

Figure 1. SSE vs. k for data.npy (blue, elbow at k = 6) and data2.npy (orange, elbow at k = 4). Dashed vertical lines mark the geometrically selected elbow for each dataset.

data.npy (blue curve). SSE falls from 605,274 at k = 1 to 71,338 at k = 5, then drops sharply to 16,710 at k = 6—a relative decrease of ~76 %. Beyond k = 6 the curve flattens, with each additional cluster yielding under 2.5 % improvement. The geometric algorithm selects k = 6, consistent with visual inspection.

data2.npy (orange curve). SSE starts at ~668,255 and declines more gradually: it falls by ~201,000 from k = 1 to k = 4 (SSE = 264,064), after which the rate of decrease slows noticeably. The geometric algorithm selects k = 4 for this dataset. However, the curve lacks the sharp visual break seen in data.npy, and the elbow is comparatively subtle—the relative SSE drop from k = 3 to k = 4 (~22 %) is modest compared to the 76 % drop seen in data.npy. This warrants corroboration with the Gap statistic [2] or silhouette analysis [4] before committing to k = 4.

(b) Structural and Statistical Assumptions of k-means

k-means optimises the within-cluster sum of squared Euclidean distances [1]. This objective encodes several assumptions:

- Spherical, isotropic clusters. The squared Euclidean distance treats all dimensions equally, implicitly assuming clusters of similar shape and variance in all directions.
- Equal cluster volumes. The Voronoi partition assigns each point to its nearest centroid; clusters with very different densities or sizes are poorly captured.

- Unimodal, convex structure. Each cluster is assumed to be contiguous and convex, meaning non-convex manifolds or ring-shaped clusters cannot be recovered.
- Fixed, known k. The number of clusters is treated as a hyperparameter rather than inferred from data.

(c) Limitations of the Elbow Method

The elbow procedure has several well-documented weaknesses, illustrated by the contrast between the two datasets in Figure 1:

- Scale sensitivity. SSE is not normalised, so the apparent sharpness of the elbow is confounded by data scale and dimensionality. Pre-processing choices (e.g., standardisation) materially alter the curve.
- Ambiguity. The gradual decline in data2.npy's SSE curve exemplifies a case where no sharp elbow exists; the geometric heuristic yields a value (k = 4) that may not reflect true cluster structure.
- No statistical baseline. The method does not compare observed SSE against a null distribution, so it cannot distinguish genuine cluster separation from random fluctuation [2].
- Single-run instability. Although k-means++ [3] with n_init = 10 mitigates local minima, the SSE curve can shift across random seeds.

(d) Methodological Improvements and Alternative Validation

Several principled alternatives exist, particularly relevant for data2.npy where the elbow is weak:

- Silhouette coefficient [4]. Measures cohesion and separation jointly, with values in [−1, 1]; the k maximising the average silhouette is preferred, and individual profiles reveal cluster quality.
- Gap statistic [2]. Compares the log-SSE of the fitted solution against its expectation under a null reference distribution. Provides a formal stopping rule and is well-suited to smooth SSE curves.
- Information-theoretic criteria. BIC or AIC applied to a Gaussian mixture model (GMM) [5] penalise model complexity, enabling direct comparison across k while accounting for parameter count.
- Stability-based selection [6]. Repeat clustering on bootstrap resamples and measure the adjusted Rand index between pairs of solutions; genuinely stable k values yield low variation.

(e) When k-means Is Inappropriate: Failure Modes and Detection

k-means is unsuitable under several well-known conditions. The weak elbow in data2.npy is itself a diagnostic signal warranting further investigation:

- Non-convex geometry (e.g., two interlocking rings, crescent shapes). Voronoi cells are convex so k-means cannot recover non-linearly separable manifolds [7]. Detection: apply spectral clustering or DBSCAN [8] alongside k-means and compare Normalised Mutual Information (NMI); large discrepancies flag geometric inadequacy.
- Heterogeneous covariance structure. When clusters differ in orientation or elongation, the spherical assumption causes misassignment at boundaries. Detection: fit a GMM with full covariance and compare BIC [5]; preference for full covariance indicates anisotropy.
- Varying density and noise. k-means greedily absorbs outliers into clusters. Detection: inspect the k-nearest-neighbour distance distribution [8]; heavy-tailed distributions suggest heterogeneous density and motivate density-based methods such as DBSCAN.
- High-dimensional spaces. Euclidean distance concentrates in high dimensions [9], dissolving cluster boundaries. Note that data2.npy has 13 dimensions vs. 9 for data.npy, increasing this risk.

Detection: measure relative contrast (max dist – min dist) / min dist; values near zero call for dimensionality reduction prior to clustering.

In summary, data.npy yields a clear, well-justified k = 6. For data2.npy, the geometric elbow suggests k = 4, but the comparatively gradual SSE decline and higher dimensionality mean this selection should be validated with the Gap statistic and silhouette analysis before substantive interpretation.

References

[1] Lloyd, S. P. (1982). Least squares quantization in PCM. IEEE Transactions on Information Theory, 28(2), 129–137.

[2] Tibshirani, R., Walther, G., & Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. Journal of the Royal Statistical Society: Series B, 63(2), 411–423.

[3] Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 1027–1035.

[4] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65.

[5] Fraley, C., & Raftery, A. E. (2002). Model-based clustering, discriminant analysis, and density estimation. Journal of the American Statistical Association, 97(458), 611–631.

[6] Ben-Hur, A., Elisseeff, A., & Guyon, I. (2002). A stability based method for discovering structure in clustered data. Pacific Symposium on Biocomputing, 7, 6–17.

[7] Ng, A. Y., Jordan, M. I., & Weiss, Y. (2001). On spectral clustering: Analysis and an algorithm. Advances in Neural Information Processing Systems (NeurIPS), 14, 849–856.

[8] Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the 2nd International Conference on Knowledge Discovery and Data Mining (KDD), 226–231.

[9] Beyer, K., Goldstein, J., Ramakrishnan, R., & Shaft, U. (1999). When is “nearest neighbor” meaningful? Proceedings of the 7th International Conference on Database Theory (ICDT), 217–235.